

Paper Title

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ABSTRACT

Your abstract goes here. A good abstract usually summarizes the research problem, the proposed approach, the main results, and the principal conclusions in a concise form.

1 INTRODUCTION

Your introduction goes here. This template provides a general starting point for writing research papers in L^AT_EX. The document structure, typography, author formatting, references, figures, tables, algorithms, theorem environments, and other common elements are handled by `template.sty`.

The examples below demonstrate several frequently used L^AT_EX features. They are intended as a quick reference and can be removed when starting a new paper.

2 SOME EXAMPLES TO GET STARTED

2.1 HOW TO CREATE SECTIONS AND SUBSECTIONS

Use `\section`, `\subsection`, and `\subsubsection` to organize the document. Numbering and formatting are handled automatically by the template.

For example:

```
\section{Method}
\subsection{Problem Formulation}
\subsubsection{Implementation Details}
```

2.2 HOW TO INCLUDE FIGURES

Figures are created with the `figure` environment. Place image files in your project directory, or in a dedicated folder such as `figures/`, and include them with `\includegraphics`.

For example, if an image is stored as `example.jpg`, it can be inserted as follows:

Figure references can be written with `\ref` or, preferably, `\cref`. For example, Figure 1 automatically includes the appropriate reference name.

LaTeX will determine a suitable float position according to the surrounding text and the placement options supplied to the environment.

2.3 HOW TO ADD TABLES

Use the `table` and `tabular` environments for tables. The template includes `booktabs`, so `\toprule`, `\midrule`, and `\bottomrule` are available.

For example, Table 1 refers to the table above.

2.4 HOW TO ADD COMMENTS

Anything following the `%` symbol on a line is treated as a comment and is not typeset in the document.

For example:



Figure 1: An example figure.

Table 1: An example table.

Method	Metric 1	Metric 2
Baseline	0.00	0.00
Proposed method	0.00	0.00

```
% This line is a comment.
This line appears in the document.
```

Comments are useful for reminders, alternative wording, implementation notes, or temporarily disabling parts of the source file.

2.5 HOW TO ADD LISTS

Numbered lists can be created with the `enumerate` environment:

1. First item.
2. Second item.

Bullet lists can be created with the `itemize` environment:

- First item.
- Second item.

2.6 HOW TO WRITE MATHEMATICS

Inline mathematics is enclosed by `$...$`. Displayed equations can be written with environments such as `equation`.

Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables with $E[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2 < \infty$. Define

$$S_n = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n} \sum_{i=1}^n X_i. \quad (1)$$

Then Equation (1) gives the sample mean.

Reusable mathematical commands and notation can be placed in `math_commands.tex` so that they can be used consistently throughout the paper.

2.7 HOW TO WRITE THEOREMS

The template provides commonly used theorem environments, including `theorem`, `property`, `proposition`, `lemma`, `corollary`, `definition`, `assumption`, `remark`, and `conjecture`.

For example:

Theorem 2.1. *For any $a, b \in \mathbb{R}$,*

$$(a + b)^2 = a^2 + 2ab + b^2.$$

Theorem environments can be referenced in the same way as equations, figures, and tables. For example, Theorem 2.1 refers to the theorem above.

Definitions, lemmas, propositions, and other theorem-like environments can be written similarly:

Definition 2.2. A sequence (x_n) is said to converge to x if, for every $\varepsilon > 0$, there exists N such that

$$|x_n - x| < \varepsilon$$

for all $n \geq N$.

For example, Theorem 2.2 gives a definition using the same numbering system.

2.8 HOW TO USE CROSS-REFERENCES

Add a `\label` to sections, equations, figures, tables, algorithms, theorem environments, and other numbered objects. They can then be referenced with `\ref` or `\cref`.

For example:

```
\section{Method}
\label{sec:method}
```

```
See \cref{sec:method}.
```

The template loads `cleveref`, which automatically inserts the appropriate object name when `\cref` or `\Cref` is used.

2.9 HOW TO ADD CITATIONS AND REFERENCES

Bibliographic entries should be stored in `references.bib`. The template uses `natbib` with numerical citations. The command `\citep` produces a numerical citation such as [3], while `\citest` includes the author name, as in Greenwade [3].

Other common bibliography types work in exactly the same way. A conference paper can be cited as [5], a book as [2], a book chapter as [1], a technical report as [4], and a preprint as [6].

Multiple references can also be cited together:

```
\citep{greenwade93, exampleconference2026, examplebook2026}
```

The corresponding BibTeX entries are stored in `references.bib`. Replace the example entries with the references used by your paper.

2.10 HOW TO USE ALGORITHMS

The template provides the `algorithm` and `algpseudocode` packages.

A minimal example is:

The algorithm can then be referenced as Algorithm 1.

Algorithm 1 Example Algorithm

Require: Input x

Ensure: Output y

- 1: Initialize y
 - 2: Update y using x
 - 3: **return** y
-

2.11 HOW TO ADJUST THE TEMPLATE

General formatting settings such as page layout, typography, title formatting, section styles, float behavior, theorem environments, and author formatting are defined in `template.sty`.

Reusable mathematical commands and notation can be placed in `math_commands.tex`.

Content specific to the paper should normally remain in `main.tex`, while bibliography entries belong in `references.bib`.

Keeping these responsibilities separate makes the template easier to reuse and maintain.

REFERENCES

- [1] Jane Doe. An example book chapter. In Alice Brown and David Wilson, editors, *Handbook of Example Research*, pages 21–40. Example Publisher, 2026. 3
- [2] John Doe. *An Example Research Book*. Example Publisher, New York, NY, 2026. 3
- [3] George D. Greenwade. The Comprehensive TeX Archive Network (CTAN). *TUGboat*, 14(3):342–351, 1993. 3
- [4] Example Research Group. An example technical report. Technical Report TR-2026-01, Example University, 2026. 3
- [5] Alice Smith and Bob Johnson. An example conference paper. In *Proceedings of the Example Conference*, pages 1–10, 2026. 3
- [6] Alice Taylor and Charlie Lee. An example preprint. arXiv preprint, 2026. 3